INDUSTRY PRACTICE PROJECT · MILITARY PRACTICE STUDENTS · 2026

FULL-STACK TASK MANAGER APPLICATION

TaskFlow

Industry Practice Project Report

Violet UI Theme · Node.js · MongoDB · JWT

Project Information

Student	Alkhan Almas
Date	June 28, 2026
Deployment	taskflow-alkhan.onrender.com
GitHub	AlmasAlkhan/Practice-work-1

TaskFlow — Industry Practice Project Report

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Project	Full-Stack Task Manager Application
Date	June 28, 2026
Design Theme	Violet / Purple UI with gradient hero banners
Local URL	http://localhost:3000
Deployment	https://taskflow-alkhan.onrender.com
GitHub	https://github.com/AlmasAlkhan/Practice-work-1

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1. Project Overview

1.1 Description

TaskFlow is a full-stack web application for managing personal and professional tasks. Users can register, log in, create color-coded categories, and manage tasks with status, priority, and due dates. The application supports search, filtering, and pagination for efficient task browsing.

1.2 Problem Statement

People often lose track of tasks across different areas of life — work, study, and personal goals. TaskFlow solves this by providing a single, secure, responsive platform where users can organize tasks into categories, track progress, and quickly find what they need using search and filters.

1.3 Tech Stack

Layer	Technology
Frontend	HTML5, CSS3, Vanilla JavaScript
Backend	Node.js, Express.js
Database	MongoDB with Mongoose ODM
Authentication	JSON Web Tokens (JWT)
Password Security	bcryptjs (12 salt rounds)
Validation	Joi
Environment Config	dotenv

1.4 Project Structure

```
task-manager/
├── backend/
│   ├── src/
│   │   ├── config/         # Database connection
│   │   ├── controllers/    # Business logic
│   │   ├── middleware/     # Auth, validation, errors
│   │   ├── models/         # Mongoose schemas
│   │   ├── routes/         # API route definitions
│   │   ├── validators/     # Joi validation schemas
│   │   └── app.js          # Express application
│   ├── server.js           # Entry point
│   ├── package.json
│   └── .env
├── frontend/
│   ├── css/styles.css
│   ├── js/                 # api, auth, app, storage, validation
│   └── index.html
├── screenshots/            # Application screenshots
├── REPORT.md
└── README.md
```

1.5 Architecture Diagram

```
┌──────────────┐    HTTP/REST     ┌──────────────┐     Mongoose     ┌──────────────┐
│   Browser    │ ◄──────────────► │   Express    │ ◄──────────────► │   MongoDB    │
│  (Frontend)  │    JSON + JWT    │  (Backend)   │                  │  (Database)  │
└──────────────┘                  └──────────────┘                  └──────────────┘
```

---

2. Setup Instructions

2.1 Prerequisites

- **Node.js** 18 or higher (`node -v`)
- **npm** (`npm -v`)
- **MongoDB** — local via Homebrew or cloud via MongoDB Atlas

2.2 Installation

```
# 1. Navigate to backend
cd task-manager/backend

# 2. Install dependencies
npm install

# 3. Configure environment
cp .env.example .env
```

2.3 Environment Variables

Variable	Description	Example
`PORT`	Server port	`3000`
`MONGODB_URI`	MongoDB connection string	`mongodb://localhost:27017/taskmanager`
`JWT_SECRET`	Secret key for JWT signing	`your_secret_key`
`JWT_EXPIRES_IN`	Token lifetime	`7d`
`NODE_ENV`	Environment mode	`development`

> **Note:** On macOS, port 5000 is often used by AirPlay Receiver. This project uses port **3000** by default.

2.4 Start MongoDB

```
brew services start mongodb-community
```

If already running, you will see: `Service mongodb-community already started`.

2.5 Run the Application

```
npm start
```

Expected output:

```
MongoDB connected
Server running on port 3000
```

Open http://localhost:3000 in your browser.

2.6 Test Account

Field	Value
Email	`test@example.com`
Password	`Password1`

Or register a new account via the Sign Up form.

---

3. Database Schema

The application uses **three related MongoDB collections** connected through references.

3.1 Entity Relationship

```
┌──────────┐          1:N          ┌─────────────┐
│   User   │ ────────────────────► │  Category   │
└──────────┘                       └─────────────┘
     │                                    │
     │ 1:N                                │ 1:N
     ▼                                    ▼
┌──────────┐                              │
│   Task   │ ◄────────────────────────────┘
└──────────┘
```

3.2 User Collection

Field	Type	Required	Description
`name`	String	Yes	Full name (min 2 chars)
`email`	String	Yes	Unique, lowercase email
`password`	String	Yes	Hashed with bcrypt (min 8 chars)
`phone`	String	No	Optional phone number
`createdAt`	Date	Auto	Registration timestamp
`updatedAt`	Date	Auto	Last update timestamp

3.3 Category Collection

Field	Type	Required	Description
`name`	String	Yes	Category name
`color`	String	Yes	Hex color code (e.g. `#7c3aed`)
`description`	String	No	Category description
`user`	ObjectId	Yes	Reference to User
`createdAt`	Date	Auto	Creation timestamp
`updatedAt`	Date	Auto	Last update timestamp

3.4 Task Collection

Field	Type	Required	Description
`title`	String	Yes	Task title
`description`	String	No	Task details
`status`	Enum	Yes	`pending` , `in_progress` , `completed`
`priority`	Enum	Yes	`low` , `medium` , `high`
`dueDate`	Date	No	Optional deadline
`user`	ObjectId	Yes	Reference to User
`category`	ObjectId	No	Reference to Category
`createdAt`	Date	Auto	Creation timestamp
`updatedAt`	Date	Auto	Last update timestamp

4. API Documentation

Base URL

```
http://localhost:3000/api
```

Authentication Header

Protected endpoints require:

```
Authorization: Bearer <jwt_token>
```

4.1 Authentication Endpoints

POST `/api/auth/register`

Register a new user.

Request:

```json
{
  "name": "John Doe",
  "email": "john@example.com",
  "password": "Password1",
  "phone": "+1 555 123 4567"
}
```

Response (201):

```json
{
  "success": true,
  "message": "Registration successful",
  "data": {
    "user": {
      "name": "John Doe",
      "email": "john@example.com",
      "phone": "+1 555 123 4567",
      "_id": "6a40f129795882c2438bf2a0"
    },
    "token": "eyJhbGciOiJIUzI1NiIs..."
  }
}
```

POST `/api/auth/login`

Request:

```
{
  "email": "test@example.com",
  "password": "Password1"
}
```

Response (200):

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "user": {
      "_id": "6a40f129795882c2438bf2a0",
      "name": "Test User",
      "email": "test@example.com"
    },
    "token": "eyJhbGciOiJIUzI1NiIs..."
  }
}
```

GET `/api/auth/profile` *(protected)*

Response (200):

```
{
  "success": true,
  "data": {
    "user": {
      "name": "Test User",
      "email": "test@example.com",
      "phone": "",
      "createdAt": "2026-06-28T10:02:17.698Z"
    },
    "stats": {
      "tasks": 1,
      "categories": 0
    }
  }
}
```

4.2 Task Endpoints *(protected)*

POST `/api/tasks`

Create a new task.

Request:

```json
{
  "title": "Complete project report",
  "description": "Write the industry practice report",
  "status": "pending",
  "priority": "high",
  "dueDate": "2026-07-01",
  "category": "64a1b2c3d4e5f6789012345"
}
```

Response (201):

```json
{
  "success": true,
  "data": {
    "title": "API Test Task",
    "description": "Created via API",
    "status": "pending",
    "priority": "high",
    "user": "6a40f129795882c2438bf2a0",
    "_id": "6a40f6395ac8ea3b139b59c7",
    "createdAt": "2026-06-28T10:23:53.939Z"
  }
}
```

GET `/api/tasks`

List tasks with **search**, **filter**, and **pagination**.

Example:

```
GET /api/tasks?search=API&status=pending&priority=high&page=1&limit=10
```

Parameter	Description
`search`	Keyword search in title and description
`status`	Filter: `pending` , `in_progress` , `completed`
`priority`	Filter: `low` , `medium` , `high`
`page`	Page number (default: 1)
`limit`	Items per page (default: 10, max: 50)

Response (200):

```json
{
  "success": true,
  "data": [
    {
      "_id": "6a40f6395ac8ea3b139b59c7",
      "title": "API Test Task",
      "description": "Created via API",
      "status": "pending",
      "priority": "high"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 10,
    "total": 1,
    "pages": 1
  }
}
```

PUT `/api/tasks/:id`

Update an existing task. Returns updated task object.

DELETE `/api/tasks/:id`

Delete a task. Returns confirmation message.

4.3 Category Endpoints *(protected)*

Method	Endpoint	Description
POST	`/api/categories`	Create category
GET	`/api/categories`	List all user categories
PUT	`/api/categories/:id`	Update category
DELETE	`/api/categories/:id`	Delete category

4.4 Health Check

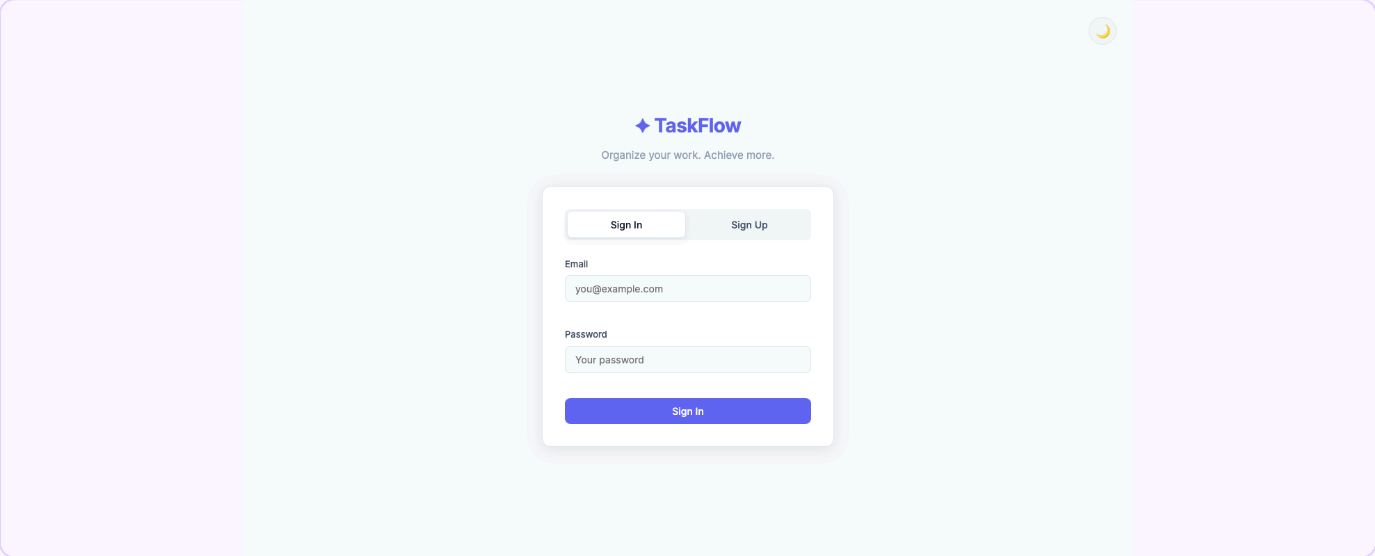
GET `/api/health`

```json
{
  "success": true,
  "message": "Task Manager API is running"
}
```

5. Key Features Screenshots

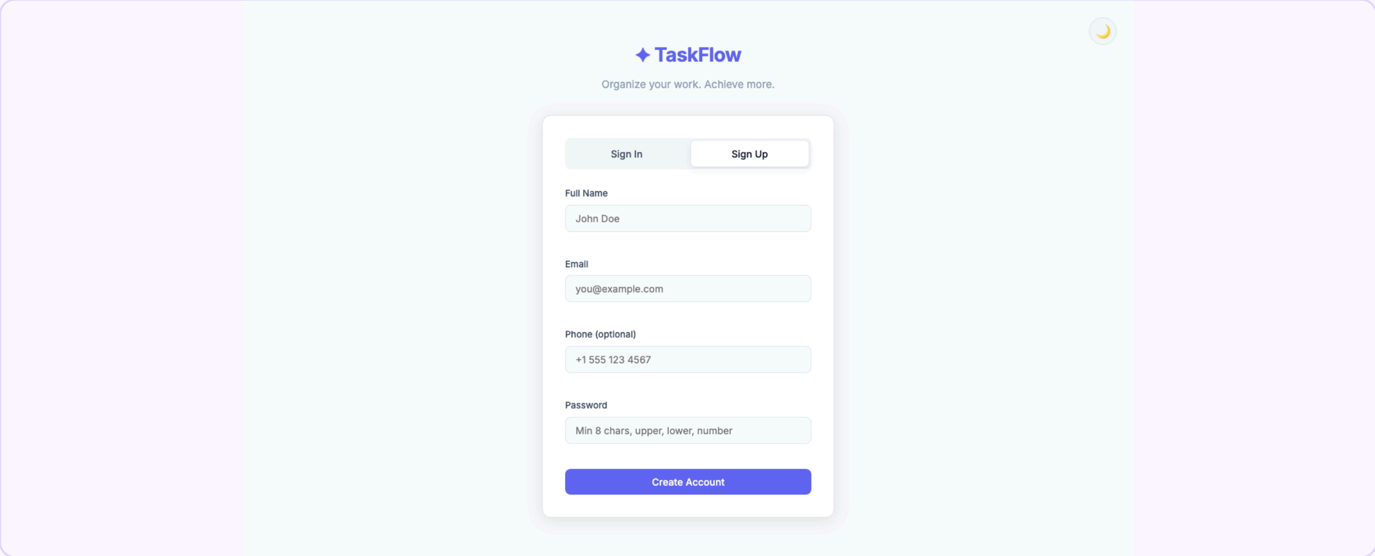
5.1 Sign In Page

Authentication form with email and password validation.



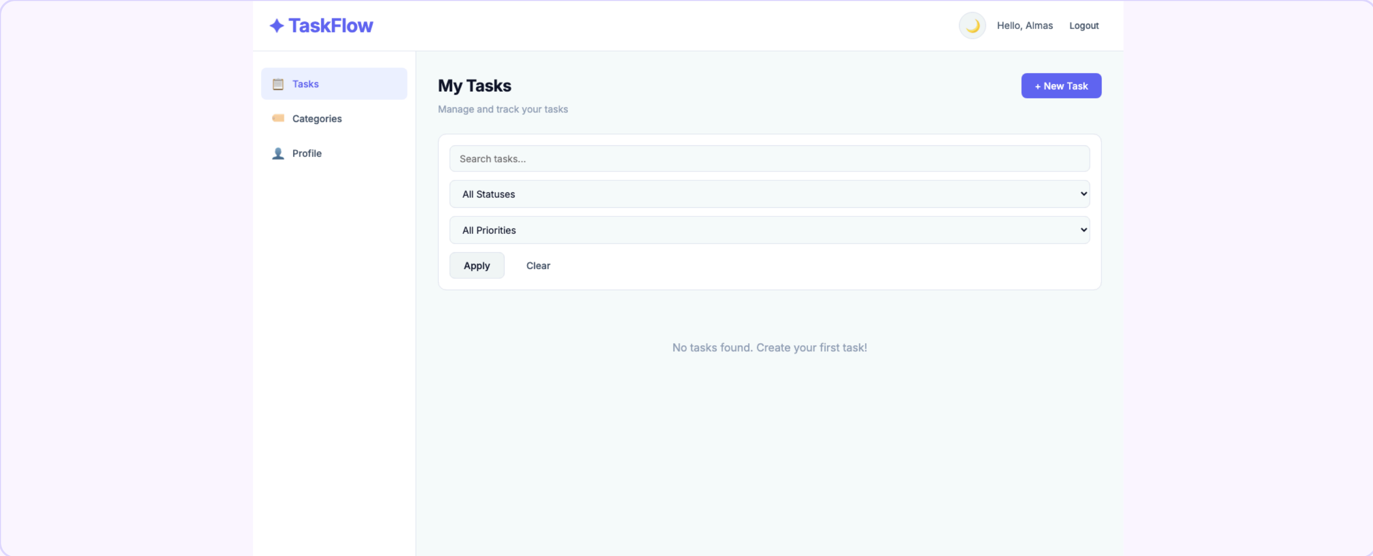
5.2 Sign Up Page

Registration form with name, email, phone, and password complexity validation.



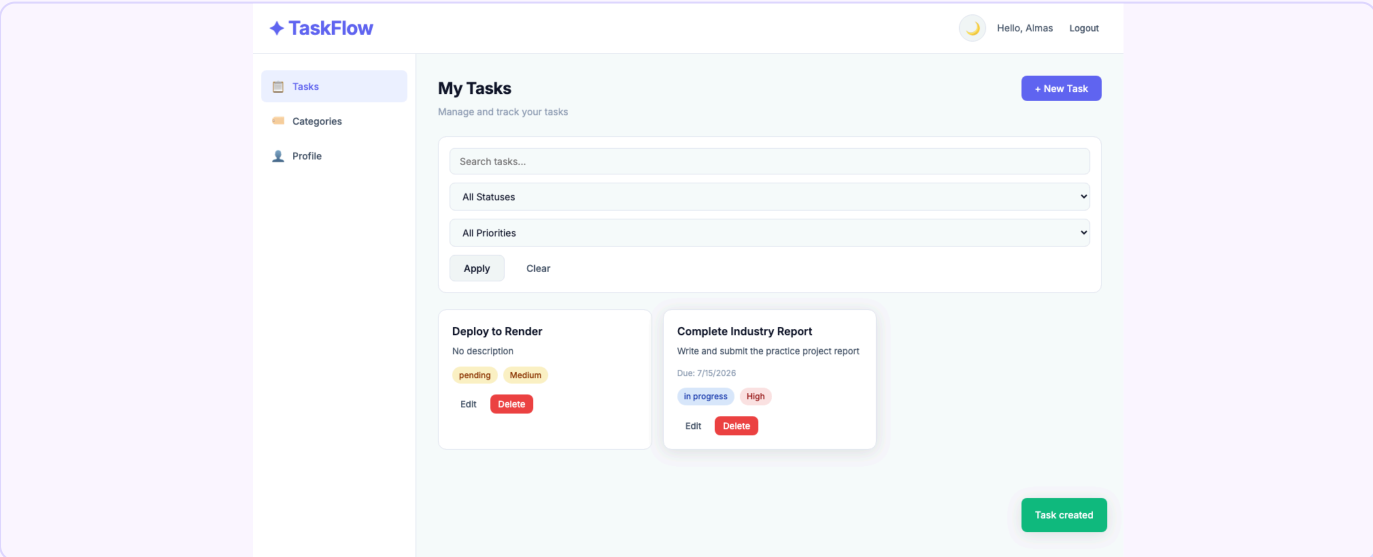
5.3 Tasks Dashboard

Main dashboard after login with sidebar navigation and task management interface.



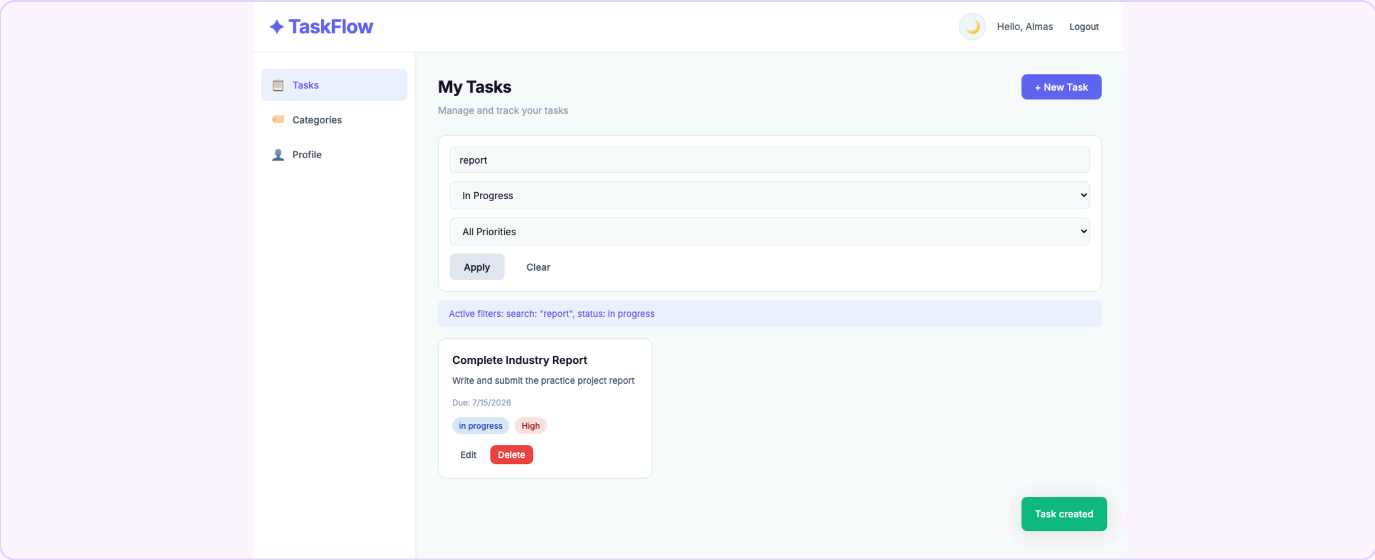
5.4 Tasks with Data

Task cards displaying title, description, status badges, priority, and due date.



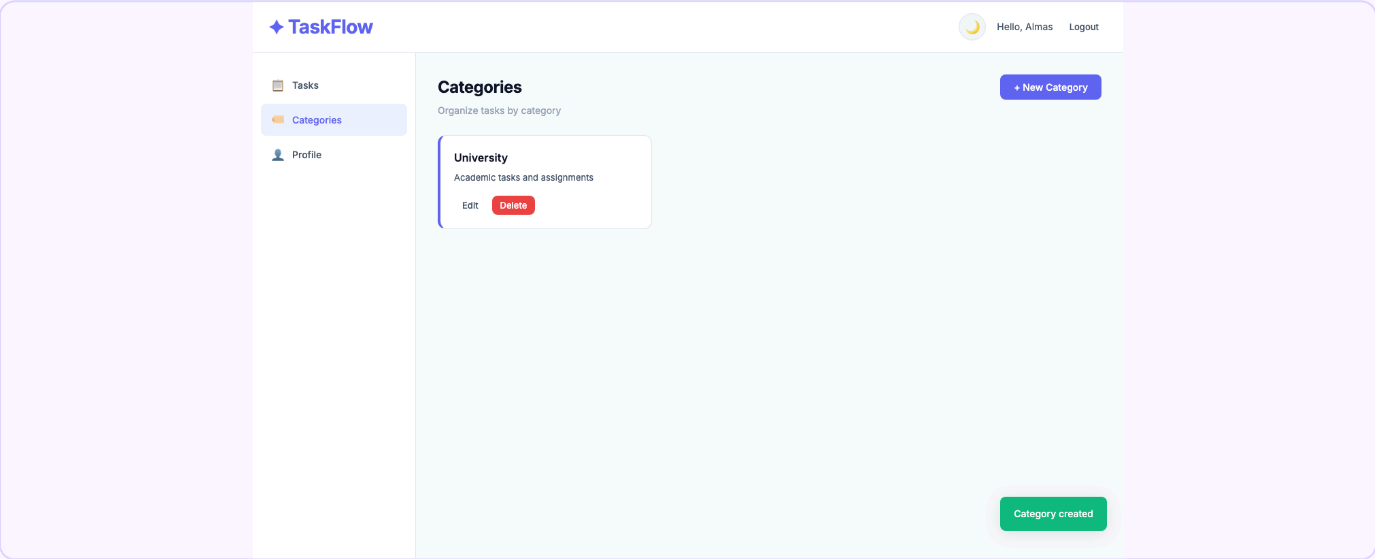
5.5 Search and Filtering

Search by keyword combined with status and priority filters. Clear button resets filters.



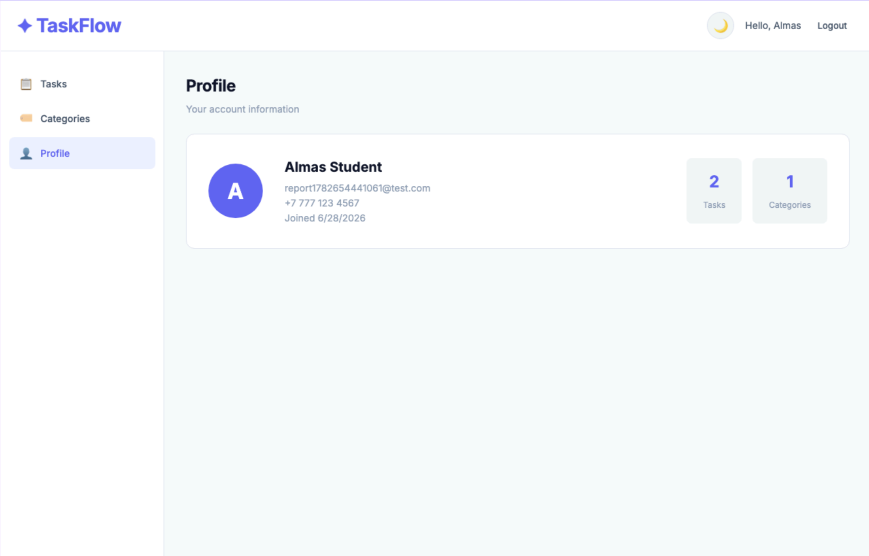
5.6 Categories Page

Color-coded category cards for organizing tasks.



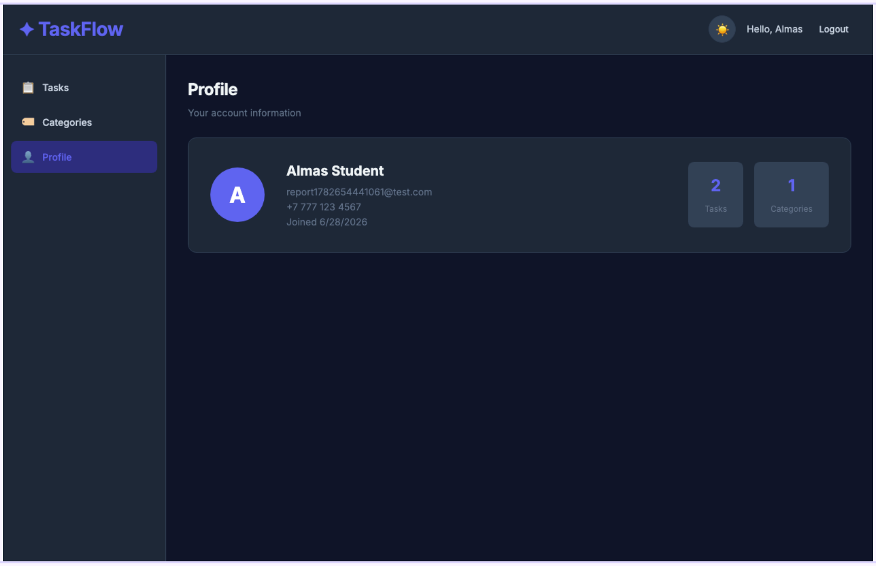
5.7 Profile Page

User account information with task and category statistics.



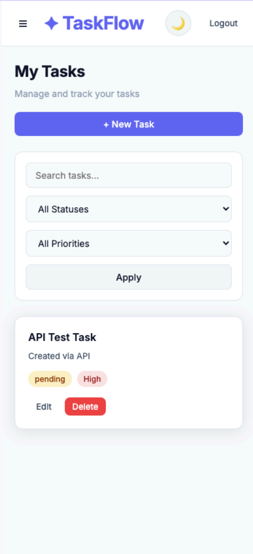
5.8 Dark Mode

Light and dark theme toggle with preference saved in localStorage.



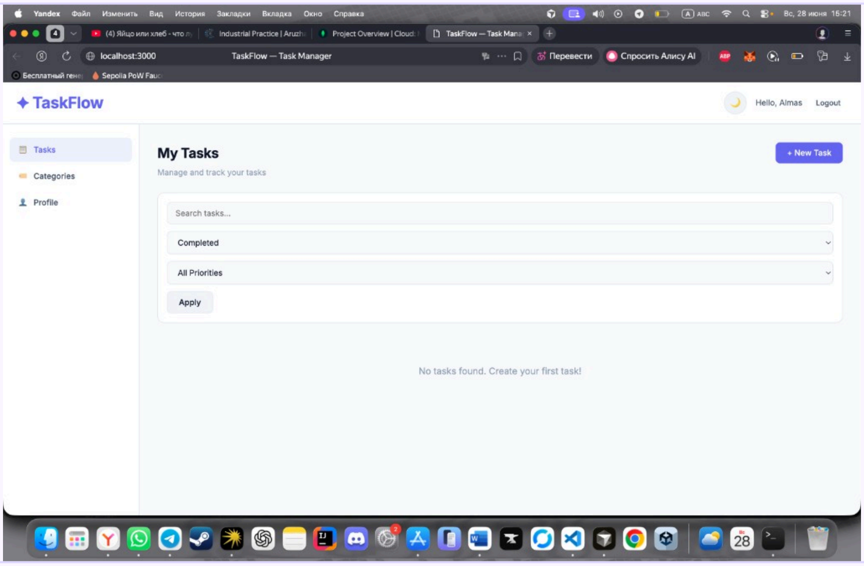
5.9 Mobile Responsive View

Fully functional layout on mobile devices (390px viewport).



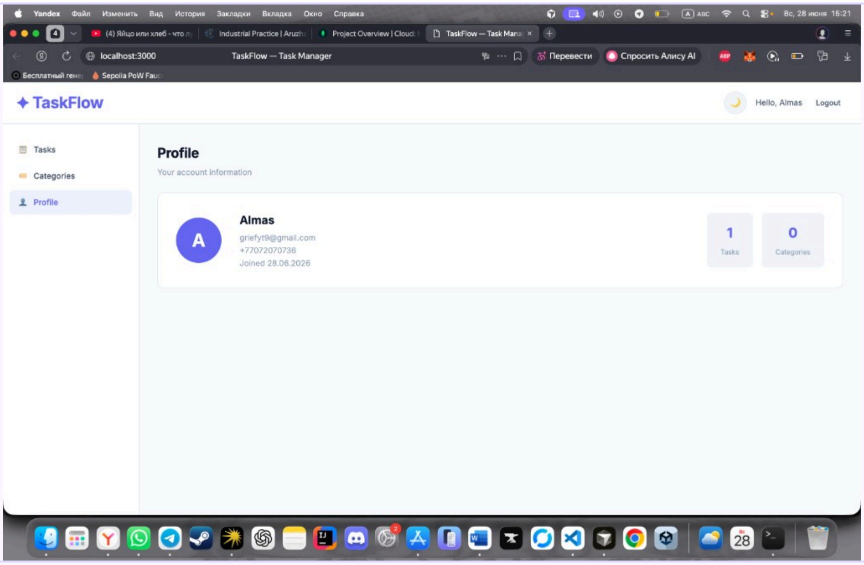
5.10 Tasks Page — Live Session

Real application screenshot from user session (Almas).



5.11 Profile Page — Live Session

Profile dashboard showing user account and statistics.



6. Authentication and Security

Feature	Implementation
JWT Authentication	Token issued on register/login, expires in 7 days
Protected Routes	`auth` middleware on `/api/tasks` and `/api/categories`
Password Hashing	bcryptjs with 12 salt rounds via Mongoose pre-save hook
Environment Variables	`JWT_SECRET`, `MONGODB_URI` stored in `.env`
CORS	Enabled for cross-origin requests
User Isolation	All queries filtered by authenticated `user` ID

Security Flow

```
1. User registers → password hashed → JWT issued
2. Client stores token in localStorage
3. Each API request includes: Authorization: Bearer <token>
4. Middleware verifies token → attaches user to request
5. Controller queries only user's own data
```

7. Validation and Error Handling

7.1 Backend Validation (Joi)

Field	Rules
Email	Valid email format
Password	Min 8 chars, uppercase + lowercase + number
Phone	Optional, 7–20 digits with `+`, spaces, dashes
Task title	Required, max 200 characters
Category name	Required, max 100 characters

7.2 Frontend Validation

- Email format check on login and registration
- Password complexity check on registration
- Phone format validation (if provided)
- Required field validation on all forms

7.3 HTTP Status Codes

Code	Usage
200	Successful GET, PUT, DELETE
201	Successful POST (created)
400	Validation error, invalid ID
401	Missing or invalid JWT token
404	Resource or route not found
409	Duplicate email on registration
500	Internal server error

7.4 Global Error Handler

Centralized `errorHandler` middleware catches:

- Mongoose validation errors → 400
- Duplicate key errors → 409
- Cast errors (invalid ObjectId) → 400
- JWT errors → 401

8. Frontend Features

8.1 Responsiveness

The application is fully responsive and tested on:

- **Desktop** — 1280px and above
- **Tablet** — 768px breakpoint with collapsible sidebar
- **Mobile** — 390px with hamburger menu navigation

8.2 Design Quality

- **Violet / purple color palette** with gradient accents (`#7c3aed` , `#9333ea` , `#c026d3`)
- Plus Jakarta Sans typography for a modern, professional look
- Top announcement bar on the login screen (Industry Practice 2026)
- Dashboard hero banner with live stats (tasks, categories, deployment status)
- Consistent color system with CSS custom properties and dark mode support
- Gradient buttons, violet-bordered task cards, and accent sidebar navigation
- Sufficient color contrast in both light and dark modes
- Status and priority badges with semantic colors
- No placeholder buttons — all UI elements are functional

8.3 User Features

Feature	Description
Sign Up	New user registration with validation
Sign In	Login with JWT token storage
Profile	Account info and statistics dashboard
Task CRUD	Create, read, update, delete tasks
Categories	Organize tasks with color-coded labels
Search	Keyword search across title and description
Filters	Filter by status and priority
Pagination	Page navigation with `page` and `limit`
Dark Mode	Theme toggle saved in localStorage
Filter Memory	Last search/filter config saved in localStorage

8.4 localStorage Usage

Key	Stored Data
`taskflow_token`	JWT authentication token
`taskflow_theme`	`light` or `dark` theme preference
`taskflow_filters`	Last search query, status, priority, page

---

9. Deployment

> **Important:** Deployment is worth **10 points**. The application is publicly accessible on Render.

9.1 Local Deployment (Development)

Item	Value
URL	http://localhost:3000
Status	✅ Running and tested

9.2 Production Deployment (Render + MongoDB Atlas)

Step 1 — MongoDB Atlas:

1. Create free M0 cluster at mongodb.com/atlas
2. Add database user with read/write permissions
3. Allow network access from `0.0.0.0/0`
4. Copy connection string

Step 2 — GitHub:

1. Push project to GitHub repository

Step 3 — Render:

1. Create new Web Service at render.com
2. Connect GitHub repository
3. Configure:

Setting	Value
Root Directory	`backend`
Build Command	`npm install`
Start Command	`npm start`

4. Set environment variables:

Variable	Value
`MONGODB_URI`	Atlas connection string
`JWT_SECRET`	Strong random secret
`JWT_EXPIRES_IN`	`7d`
`NODE_ENV`	`production`

Deployment URL:

```
https://taskflow-alkhan.onrender.com
```

GitHub Repository:

```
https://github.com/AlmasAlkhan/Practice-work-1
```

Author: **Alkhan Almas**

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10. Project Checklist

#	Requirement	Points	Status
1	Project setup with modular structure	10	✅
2	Database with 3 related entities	10	✅
3	API endpoints (auth, CRUD, search/filter/pagination)	20	✅
4	JWT, bcrypt, protected routes, .env	15	✅
5	Joi validation, error handling, form validation	10	✅
6	Responsive design, professional UI, dark mode	15	✅
7	Sign Up, Login, Profile, search, localStorage	10	✅
8	Deployment with public URL	10	✅ https://taskflow-alkhan.onrender.com
	Report with overview, setup, API docs, screenshots		✅ Done

Conclusion

TaskFlow is a complete full-stack CRUD application developed by **Alkhan Almas** that meets all technical requirements of the Industry Practice Project. The application is deployed at **https://taskflow-**

alkhan.onrender.com and the source code is available at https://github.com/AlmasAlkhan/Practice-work-1.

Report prepared by Alkhan Almas — Industry Practice Project, Military Practice Students